

Text 2: Challenging medical thinking on placebos

Dr Damien Finniss, Associate Professor at Sydney University's Pain Management and Research Institute, was previously a physiotherapist. He regularly treated football players during training sessions using therapeutic ultrasound. 'One particular session', Finniss explains, 'I treated five or six athletes. I'd treat them for five or ten minutes and they'd say, "I feel much better" and run back onto the field. But at the end of the session, I realised the ultrasound wasn't on.' It was a light bulb moment that set Finniss on the path to becoming a leading researcher on the placebo effect.

Used to treat depression, psoriasis and Parkinson's, to name but a few, placebos have an image problem among medics. For years, the thinking has been that a placebo is useless unless the doctor convinces the patient that it's a genuine treatment – problematic for a profession that promotes informed consent. However, a new study casts doubt on this assumption and, along with a swathe of research showing some remarkable results with placebos, raises questions about whether they should now enter the mainstream as legitimate prescription items. The study examined five trials in which participants were told they were getting a placebo, and the conclusion was that doing so honestly can work.

'If the evidence is there, I don't see the harm in openly administering a placebo,' says Ben Colagiuri, a researcher at the University of Sydney. Colagiuri recently published a meta-analysis of thirteen studies which concluded that placebo sleeping pills, whose genuine counterparts **notch up** nearly three million prescriptions in Australia annually, significantly improve sleep quality. The use of placebos could therefore reduce medical costs and the burden of disease in terms of adverse reactions.

But the placebo effect isn't just about fake treatments. It's about raising patients' expectations of a positive result; something which also occurs with real drugs. Finniss cites the 'open-hidden' effect, whereby an analgesic can be twice as effective if the patient knows they're getting it, compared to receiving it unknowingly. 'Treatment is always part medical and part ritual,' says Finniss. This includes the austere consulting room and even the doctor's clothing. But behind the performance of healing is some strong science. Simply believing an analgesic will work activates the same brain regions as the genuine drug. 'Part of the outcome of what we do is the way we interact with patients,' says Finniss.

That interaction is also the focus of Colagiuri's research. He's looking into the 'nocebo' effect, when a patient's pessimism about a treatment becomes self-fulfilling. 'If you give a placebo, and warn only 50% of the patients about side effects, those you warn report more side effects,' says Colagiuri. He's aiming to reverse that by exploiting the psychology of food packaging. Products are labelled '98% fat-free' rather than '2% fat' because positive reference to the word 'fat' puts consumers off. Colagiuri is deploying similar tactics. A drug with a 30% chance of causing a side effect can be reframed as having a 70% chance of not causing it. 'You're giving the same information, but framing it a way that minimises negative expectations,' says Colagiuri.

There is also a body of research showing that a placebo can produce a genuine biological response that could affect the disease process itself. It can be traced back to a study from the 1970s, when psychologist Robert Ader was trying to condition taste-aversion in rats. He gave them a saccharine drink whilst simultaneously injecting Cytoxan, an immune-suppressant which causes nausea. The rats learned to hate the drink due to the nausea. But as Ader continued giving it to them, without Cytoxan, they began to die from infection. Their immune system had 'learned' to fail by repeated pairing of the drink with Cytoxan. Professor Andrea Evers of Leiden University is running a study that capitalises on this conditioning effect and may benefit patients with rheumatoid arthritis, which causes the immune system to attack the joints. Evers' patients are given the immunosuppressant methotrexate, but instead of always receiving the same dose, they get a higher dose followed by a lower one. The theory is that the higher dose will cause the body to link the medication with a damped-down immune system. The lower dose will then work because the body has 'learned' to curb immunity as a placebo response to taking the drug. Evers hopes it will mean effective drug regimes that use lower doses with fewer side effects.

The medical profession, however, remains less than enthusiastic about placebos. 'I'm one of two researchers in the country who speak on placebos, and I've been invited to lecture at just one university,' says Finniss. According to Charlotte Blease, a philosopher of science, this antipathy may go to the core of what it means to be a doctor. 'Medical education is largely about biomedical facts. 'Softer' sciences, such as psychology, get marginalised because it's the hard stuff that's associated with what it means to be a doctor.' The result, says Blease, is a large, placebo-shaped hole in the medical curriculum. 'There's a great deal of medical illiteracy about the placebo effect ... it's the science behind the art of medicine. Doctors need training in that.'

15. A football training session sparked Dr Finniss' interest in the placebo effect because
- (A) he saw for himself how it could work in practice.
 - (B) he took the opportunity to try out a theory about it.
 - (C) he made a discovery about how it works with groups.
 - (D) he realised he was more interested in research than treatment.
16. The writer suggests that doctors should be more willing to prescribe placebos now because
- (A) research indicates that they are effective even without deceit.
 - (B) recent studies are more reliable than those conducted in the past.
 - (C) they have been accepted as a treatment by many in the profession.
 - (D) they have been shown to relieve symptoms in a wide range of conditions.
17. What is suggested about sleeping pills by the use of the verb 'notch up'?
- (A) they may have negative results
 - (B) they could easily be replaced
 - (C) they are extremely effective
 - (D) they are very widely used
18. What point does the writer make in the fourth paragraph?
- (A) The way a treatment is presented is significant even if it is a placebo.
 - (B) The method by which a drug is administered is more important than its content.
 - (C) The theatrical side of medicine should not be allowed to detract from the science.
 - (D) The outcome of a placebo treatment is affected by whether the doctor believes in it.

19. In researching side effects, Colagiuri aims to
- (A) discover whether placebos can cause them.
 - (B) reduce the number of people who experience them.
 - (C) make information about them more accessible to patients.
 - (D) investigate whether pessimistic patients are more likely to suffer from them.
20. What does the word 'it' in the sixth paragraph refer to?
- (A) a placebo treatment
 - (B) the disease process itself
 - (C) a growing body of research
 - (D) a genuine biological response
21. What does the writer tell us about Ader's and Evers' studies?
- (A) Both involve gradually reducing the dosage of a drug.
 - (B) Evers is exploiting a response which Ader discovered by chance.
 - (C) Both examine the side effects caused by immunosuppressant drugs.
 - (D) Evers is investigating whether the human immune system reacts to placebos as Ader's rats did.
22. According to Charlotte Blease, placebos are omitted from medical training because
- (A) there are so many practical subjects which need to be covered.
 - (B) those who train doctors do not believe that they work.
 - (C) they can be administered without specialist training.
 - (D) their effect is more psychological than physical.